Year 10 Higher

REVISION PACK

Number 3

Name: _____

Class: _____

Number 3 · Negative and fractional indices (Lessons 6.1, 6.2)

Worked example: working out the value of powers

Work out the value of

(a) 4^{-3} (b) $8^{\frac{2}{3}}$ (c) $\left(\frac{25}{4}\right)^{-\frac{1}{2}}$

(a)	**Step 1**	Minus power: 1 over	$4^{-3} = \frac{1}{4^3}$
	Step 2	Answer	$\frac{1}{64}$
(b)	**Step 1**	Bottom 3: cube root	$\sqrt[3]{8} = 2$
	Step 2	Top 2: square	$2^2 = 4$
	Step 3	Answer	4
(c)	**Step 1**	Minus: flip	$\left(\frac{4}{25}\right)^{\frac{1}{2}}$
	Step 2	Square root top and bottom	$\frac{\sqrt{4}}{\sqrt{25}}$
	Step 3	Answer	$\frac{2}{5}$

Practice: working out the value of powers

1.1 Work out the value of

(a) 3^{-2} (b) $27^{\frac{2}{3}}$ (c) $\left(\frac{64}{27}\right)^{-\frac{1}{3}}$

1.2 Work out the value of

(a) 2^{-5} (b) $16^{\frac{3}{2}}$ (c) $\left(\frac{9}{16}\right)^{-\frac{1}{2}}$

1.3 Work out the value of

(a) 10^{-3} (b) $125^{\frac{2}{3}}$ (c) $\left(\frac{8}{125}\right)^{-\frac{1}{3}}$

1.4 Work out the value of

(a) 4^{-2} (b) $9^{\frac{3}{2}}$ (c) $\left(\frac{25}{36}\right)^{-\frac{1}{2}}$

1.5 Work out the value of

(a) 5^{-3} (b) $8^{\frac{4}{3}}$ (c) $\left(\frac{27}{64}\right)^{-\frac{1}{3}}$

1.6 Work out the value of

(a) 2^{-6} (b) $100^{\frac{3}{2}}$ (c) $\left(\frac{49}{4}\right)^{-\frac{1}{2}}$

Number 3 · Negative and fractional indices (Lessons 6.1, 6.2)

Worked example: finding the power n

Find the value of n.

(a) $\frac{1}{49} = 7^n$ (b) $2^n \div 2^6 = 2^{-2}$

(a)	**Step 1**	Bottom as a power	$49 = 7^2$
	Step 2	1 over: minus power	$\frac{1}{7^2} = 7^{-2}$
	Step 3	Answer	$n = -2$
(b)	**Step 1**	Divide: subtract the powers	$n - 6 = -2$
	Step 2	Add 6	$n = -2 + 6$
	Step 3	Answer	$n = 4$

Practice: finding the power n

1.7 Find the value of n.

(a) $\frac{1}{16} = 2^n$ (b) $5^n \div 5^3 = 5^{-1}$

1.8 Find the value of n.

(a) $\frac{1}{27} = 3^n$ (b) $4^n \div 4^5 = 4^{-2}$

1.9 Find the value of n.

(a) $\frac{1}{100} = 10^n$ (b) $6^n \div 6^7 = 6^{-3}$

1.10 Find the value of n.

(a) $\frac{1}{64} = 4^n$ (b) $2^n \div 2^9 = 2^{-4}$

1.11 Find the value of n.

(a) $\frac{1}{81} = 9^n$ (b) $7^n \div 7^3 = 7^{-2}$

1.12 Find the value of n.

(a) $\frac{1}{8} = 2^n$ (b) $10^n \div 10^6 = 10^{-1}$